

A 2D Structure-from-Motion Pipeline for Sparse Planar Map Reconstruction with DLT, RANSAC, and Bundle Adjustment

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Abstract—This report describes a 2D Structure-from-Motion pipeline that reconstructs a sparse planar map from a set of images capturing the same scene under varying viewpoints. The core problem is to estimate a homography for each image that maps its content into a shared global coordinate frame, from which all keypoints can be jointly visualized as a 2D point cloud. The base pipeline extracts SIFT keypoints and descriptors from each image, matches them pairwise using a brute-force KNN matcher with Lowe’s ratio test, and estimates pairwise homographies via the Direct Linear Transform (DLT) with RANSAC. A greedy motion extension algorithm initializes the reconstruction from the most confidently matched image pair based on inlier count and chains further images in by composing pairwise homographies. The advanced pipeline extends this with bundle adjustment: feature tracks are constructed across all images using a union-find data structure, and the global keypoint positions and all non-reference homographies are jointly refined by minimizing the reprojection error via mini-batch stochastic Gauss-Newton optimization. The pipeline is evaluated on four image datasets called bark, boat, graf, and wall using mean Euclidean distance between estimated and ground-truth keypoints as the quality metric.

Index Terms—structure-from-motion, homography, DLT, RANSAC, bundle adjustment, SIFT, feature tracks

I. METHOD

A. DLT With RANSAC Homography Estimation

The following sections describe the pipeline components which together estimate the homographies H_I which bring points from image I to the reference frame for an image J such that a 2D planar sparse map reconstruction of the image contents is created from the set of images $\{I_1, I_2, \dots, I_N\}$ which may have differing perspective as well as reference points for the image contents. Algorithm 1 summarizes the complete pipeline, with each step described in detail in the following subsections.

1) *Feature Extraction*: To begin, SIFT features, keypoints and descriptors, are extracted from each image in the dataset. For each unique unordered pair of images the descriptors are matched via a brute-force KNN matcher with $k = 2$ using the Euclidean norm as a distance metric. The Lowe’s ratio test is applied to each match to reduce the number of false matches. The ratio test only accepts matches where the ratio

Algorithm 1 2D Structure-from-Motion Pipeline

Require: Image set $\{I_1, \dots, I_N\}$; RANSAC threshold τ ; iterations n_{iter}
Ensure: Global homographies $\{H_I\}$ and sparse 2D map $\mathcal{M}$

- 1: **for all** images I **do**
- 2: Extract SIFT keypoints and descriptors
- 3: **end for**
- 4: **for all** unordered image pairs (I, J) **do**
- 5: Match descriptors (brute-force KNN, $k = 2$) with Lowe’s ratio test
- 6: **if** number of matches ≥ 4 **then**
- 7: Normalize points of I and J via similarity transforms T, T'
- 8: $(H_{J,I}, \text{inliers}) \leftarrow \text{RANSAC} - \text{DLT}(\text{matches}, \tau, n_{\text{iter}})$
- 9: Store $H_{J,I}$ and its inlier count
- 10: **end if**
- 11: **end for**
- 12: $(I, J) \leftarrow$ pair with the largest inlier count
- 13: $H_J \leftarrow \text{Id}$; $H_I \leftarrow H_{J,I}$ $\{J$ defines the global frame $\}$
- 14: Mark I, J as included
- 15: **while** images remain unincluded **do**
- 16: Select unincluded image I with most inliers to any included image J
- 17: $H_I \leftarrow H_J \cdot H_{J,I}$
- 18: Mark I as included
- 19: **end while**
- 20: $\mathcal{M} \leftarrow \bigcup_I \{H_I \mathbf{x} : \mathbf{x} \in \text{keypoints}(I)\}$ {homogeneous division applied}
- 21: (*Advanced*) Refine $\{H_I\}$ and global points by bundle adjustment (Alg. 2); regenerate $\mathcal{M}$
- 22: **return** $\{H_I\}, \mathcal{M}$

between the first and second nearest neighbor is less than a threshold value. In this case, using a threshold of 0.8, we only keep those matches where the nearest neighbor is at least 20% closer than the second nearest neighbor.

2) *Homography Estimation*: For each unique unordered pair of images with at least 4 matches, the homography H that satisfies $\mathbf{x}' = H\mathbf{x}$ is estimated using the Direct Linear Transform (DLT) with RANSAC.

a) *Normalization*: Keypoint normalization is independently applied to the point sets of both images before applying the DLT algorithm. This process involves calculating a similarity transform $\mathbf{T}$ which centers the points around the origin and scales points to an average distance of $\sqrt{2}$ from the origin. Normalization of this form aims to rescale pixel values to a range which improves the numerical conditioning of the DLT algorithm. This mitigates the potential for poor conditioning introduced by noisy measurements with large value differences which create greatly amplified errors in the estimation. At worst the DLT algorithm can fail to produce a result because of poor conditioning.

b) *DLT*: For each point correspondence $(\mathbf{x}_i, \mathbf{x}'_i)$, the cross-product constraint $\mathbf{x}'_i \times H\mathbf{x}_i = 0$ yields two linear equations in the 9 entries of H , assembled vertically into a $2n \times 9$ matrix A [1]. However, in this implementation only four point correspondences are ever used yielding an 8×9 matrix A . The solution to H is then given by solving the null-space of the equation $A\mathbf{h} = 0$ where $\mathbf{h} = (\mathbf{h}^1 \ \mathbf{h}^2 \ \mathbf{h}^3)^T$ and $\mathbf{h}^i$ is one row of H stacked vertically. However, since in practice measurements are noisy, the exact null-space solution does not exist. Therefore, the approximate null-space solution must be found which is given by the (unit) eigenvector of $A^T A$ with least eigenvalue. Equivalently, the singular vector is obtained by singular value decomposition (SVD) which factorizes any real or complex matrix B into the matrix product $B = U\Sigma V^T$. The factor Σ is a diagonal matrix with singular values σ_i in descending order along the diagonal. Consequently, we can take the last column of V as our solution to $\mathbf{h}$. Finally, the denormalized homography is recovered as $H = T'^{-1}\tilde{H}T$.

c) *RANSAC*: Random selection of four point correspondences constitutes the minimum number of measurements required to solve the minimal problem of estimating a homography between them. This property is exploited by the RANSAC algorithm which iteratively selects four random keypoint matches between the image pair and applies the DLT algorithm to estimate the homography H . Using the estimated homography the reprojection error of the keypoints is quantified by the residual sum of squares. Using a prefixed threshold on this residual, the number of inliers among the keypoints is determined and recorded. RANSAC performs this process for a set number of iterations and returns the estimated homography between the two images with the highest number of inliers. This process is used instead of a direct DLT transformation on all point correspondences in order to mitigate estimation of poor or false homographies based on outlier matches and noisy data.

3) *Extend Motion*: For each unique unordered pair with a homography $\mathbf{H}$ estimated from RANSAC via DLT, the pair with the largest inlier count is selected as the starting pair (I, J) for the reconstruction. Image J defines the global coordinate system (identity homography), and image I 's global homography is set to $H_{J,I}$ (mapping $I \rightarrow J$). Following this initialization, further images are added greedily from the remaining available pairs. At each extend motion step, among all images not yet in the reconstruction, the image I with the highest inlier count on the homography estimation to any already-included image J is selected. The global homography for I is then calculated as $H_I = H_J \cdot H_{J,I}$ where H_J is the already-known global homography for J . This process repeats until there are no more images to select from.

4) *Sparse Map Reconstruction*: After calculating the homography H_I which describes the transform from image I to the global coordinate system determined by the starting pair, all SIFT keypoints from each image I are mapped into the global coordinate system with homogeneous division applied after. The union of all mapped keypoints forms the sparse 2D map.

5) *Evaluation*: The main quality metric for the evaluation of the implemented sparse map reconstruction pipeline is the mean Euclidean distance, measured in pixels, between corresponding estimated and ground-truth keypoints in all images for a given dataset. In order to calculate the residuals correctly we must first transform the ground-truth keypoints into the same reference frame as the estimated keypoints. To this end we iteratively transform all keypoints in each image with the inverse of its respective ground-truth homography to the reference frame of the first image, as these are the ground-truth homographies available to us. Then the homography transforming from the first image to the correct reference image is applied. Finally, the residuals are calculated between these two keypoint correspondences and the average is taken.

B. Bundle Adjusted DLT With RANSAC Homography Estimation

The following sections describe the pipeline components which extend the process described in Section I-A with the additional extension of bundle adjustment (BA) which jointly refines the reconstruction homographies across all images using Gauss-Newton optimization. This process improves the final result by estimating the homographies for each image globally as a connected set of images and points instead of estimating each pairwise homography in isolation. Algorithm 2 outlines the bundle adjustment procedure, detailed in the subsections below.

1) *Feature Tracks and Global Points*: The main data structure converting the isolated pairwise homography estimation to a joint estimation is feature tracks. Consider the transitive property of the keypoint matching relation R between images where if an image and keypoint pair $(I^i, \text{kp}^{i'})$ matches $(I^j, \text{kp}^{j'})$ and $(I^j, \text{kp}^{j'})$ matches $(I^k, \text{kp}^{k'})$ then $(I^i, \text{kp}^{i'})$ matches $(I^k, \text{kp}^{k'})$. Consequently, we can define a feature track as the set of image and keypoint pairs that form a

Algorithm 2 Bundle Adjustment (mini-batch Gauss-Newton)**Require:** Initial $\{H_I\}$, feature tracks, step size α , sample ratio ρ **Ensure:** Refined parameter vector θ

- 1: Build feature tracks via union-find over inlier correspondences
- 2: Initialize each global point as the mean of its transformed observations
- 3: Pack global points and non-reference homographies ($h_{3,3} = 1$ fixed) into θ
- 4: Sample a fixed held-out track subset for early-stopping evaluation
- 5: **repeat**
- 6: Sample a random subset of tracks (ratio ρ , clamped to $[50, 500]$)
- 7: Compute residuals $\mathbf{r}$ and finite-difference Jacobian J on the subset
- 8: Solve $J\delta = \mathbf{r}$ (least-squares); $\theta \leftarrow \theta - \alpha \delta$
- 9: Evaluate mean $\|\mathbf{r}\|$ on held-out subset; track best θ
- 10: **until** max iterations reached **or** $\|\delta\| < \epsilon$ **or** patience exceeded
- 11: **return** best θ

sequence of transitive relations. Together with the definition of our matching relation R , the transitive property implies that each element in the feature track corresponds to the same global point of image content. Therefore, we may define the global point position for each feature track as the average of each keypoint pair transformed to the global (reference) frame via the RANSAC and DLT estimated homographies H_I .

A union-find data structure [2] which stores a collection of disjoint sets with operations for adding, merging, and finding is used to determine the feature tracks among all images. The union-find data structure is based on a forest structure where each disjoint set is considered its own tree where each element in the collection is tied to a specific set (tree) by storing a pointer to the root element of the set. Consider the pairs $(I^j, \text{kp}^{j'})$ and $(I^k, \text{kp}^{k'})$ from the previous example being added to the data structure by a union operation. Then, for each pair, a traversal up the tree is performed to determine the root, denoted as a find operation, of the pair. If a root is found, for example the matching pairs $(I^i, \text{kp}^{i'})$ and $(I^j, \text{kp}^{j'})$ were previously added, then the find operation returns the representative root of the set already containing $(I^j, \text{kp}^{j'})$. If $(I^k, \text{kp}^{k'})$ has not previously been seen, the find operation initializes it as its own root with rank 0. The union operation then compares the ranks of the two roots: the root of the set $(I^i, \text{kp}^{i'})$, $(I^j, \text{kp}^{j'})$ has rank 1 (incremented when those two rank-0 singletons were merged), while $(I^k, \text{kp}^{k'})$ has rank 0. Since the existing set has higher rank, $(I^k, \text{kp}^{k'})$ is attached under that root, merging all three observations into a single track.

To extract the feature tracks from the union-find data structure we simply iterate over all disjoint sets and place their

elements into a list.

2) *Parameter Vector Construction and Deconstruction:* The parameters we wish to optimize over in BA are denoted as θ where the first $2M$ entries consist of the x - and y -coordinates flattened row-major of the M global keypoints calculated from the feature tracks. The remaining entries of θ consist of the first 8 elements of each flattened row-major non-reference homography. In this construction we have implicitly set $h_{3,3} = 1$ of each H_I to remove scale ambiguity. Furthermore, the reference homography which is the identity matrix is not included as it defines the reference frame used for the global points. Adding it to θ makes the reference parameters optimizable, which in turn renders the optimization of the other parameters meaningless by drifting to some optimum not related to our reference frame.

Deconstruction of θ is done by rearranging the M global points back into a $(2, N)$ shape and the homographies are similarly reconstructed block-wise with an additional $h_{3,3} = 1$ inserted.

3) *Computation of Residuals:* For each observation $(I^i, \text{kp}^{i'})$ in each track, the reprojection of the track's optimized global point into that image's coordinate frame is computed by applying the inverse of the optimized image's homography. The residual is the 2D difference between the observed keypoint position and this reprojection. All residuals across all observations are stacked into a vector $\mathbf{r}$.

4) *Computation of Jacobian:* Computing the full Jacobian for this problem is costly and not reasonable on the hardware or methods available within this project. Therefore, the Jacobian is calculated via finite difference approximation. The Jacobian J of the residual vector $\mathbf{r}$ with respect to θ is then defined as

$$J_k = \frac{\partial \mathbf{r}}{\partial \theta_k} \approx \frac{\mathbf{r}(\theta_k + \varepsilon) - \mathbf{r}(\theta_k)}{\varepsilon}, \quad (1)$$

where J_k is the k -th column of J .

5) *Gauss-Newton Update:* Due to the number of parameters, observed keypoints in feature tracks, and the cost of calculations, it is unreasonable to perform a Gauss-Newton update over all of the available data due to time constraints. Therefore, mini-batch stochastic Gauss-Newton is used instead at the cost of a noisier update. This is accomplished by sampling a random subset of feature tracks according to a provided ratio (with a minimum of 50 and maximum of 500 tracks allowed) and constructing J and $\mathbf{r}$ from the subset. The update is then defined as

$$\theta^{n+1} = \theta^n - \alpha(J^T J)^{-1} J^T \mathbf{r}, \quad (2)$$

which in the implementation is realized by solving the linear least-squares problem $J\delta = \mathbf{r}$ where $\delta = (J^T J)^{-1} J^T \mathbf{r}$ using `np.linalg.lstsq` and $\alpha = 0.05$ as a fixed step-size.

The optimization can terminate on three different conditions, firstly by reaching the maximum number of iterations provided as a parameter, secondly by the norm of δ being smaller than a given threshold, and finally by early stopping with a fixed patience of ten. Early stopping is implemented by

calculating the mean of the absolute residual vector $|\mathbf{r}|$ from the updated parameter vector on a held-out predetermined subset of the feature tracks. This mean is compared to the best recorded and if ten iterations pass without improvement the optimization loop is broken and the best θ is returned.

6) *Reconstruction Update*: After optimization has completed, the optimized homographies are unpacked from θ and written back into the reconstruction, replacing the original RANSAC-estimated homographies. The sparse map is then regenerated by applying these optimized homographies to all SIFT keypoints in each image, exactly as in the basic pipeline.

II. EXPERIMENTAL EVALUATION

This section details the experiments and results of evaluating both implemented methods.

A. DLT With RANSAC

In order to be as comprehensive as possible in the experimental evaluation of the implemented pipeline, a grid-search across the hyper-parameters to the RANSAC algorithm was performed where the average residual between the ground truth points and the structure from motion estimated points was used as a quality metric. The results of this hyperparameter search are shown in Appendix B while the best results are summarized in Table I.

TABLE I
SUMMARY RESULTS OF THE DLT WITH RANSAC METHOD GRID SEARCH.

dataset	thresh	n_iter	avg_residual
bark	0.001	100	2.9697
boat	1e-05	1000	4.7630
graf	1e-05	1000	1.2388
wall	1e-05	100	1.5678

The best result for each dataset is shown in Fig. 1, Fig. 2, Fig. 3, and Fig. 4 where it is clear in all cases that the estimated structure from motion point cloud matches the ground-truth point cloud within a very low error.

Another method of evaluation performed was overlaying the ground-truth and structure from motion estimated point clouds on the reference image itself to observe if the keypoints match the image contents. The scatter plot is cropped to the image width and height in order to focus on the points relevant to the reference image more clearly. These results are presented in Fig. 5, Fig. 6, Fig. 7, and Fig. 8. The quality of the estimation is clear from the absence of any difference in the overlays of the image.

The worst result for each dataset is shown in Fig. 9, Fig. 10, Fig. 11, and Fig. 12 where it is clear in all cases that the estimated structure from motion point cloud does not match the ground-truth point cloud at all.

Specifically, we can observe that the structure from motion estimations fail due to large outlier values being projected into the reference coordinate frame. This coincides with the hyperparameters used to create them since all poor estimations

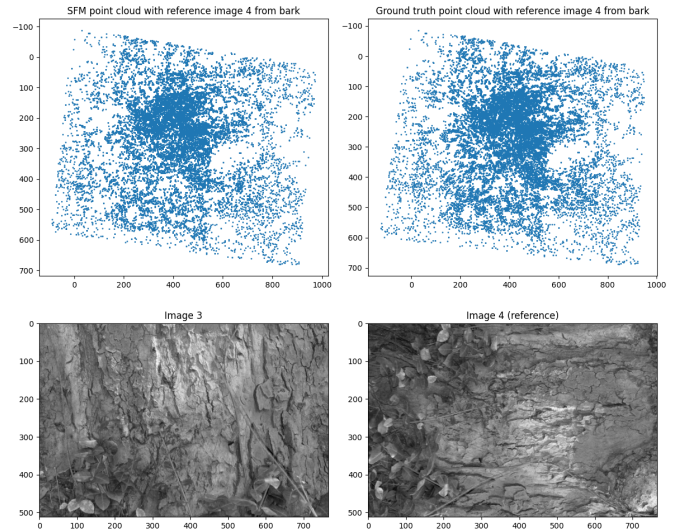


Fig. 1. Structure from motion estimated point cloud with parameters $\text{thresh} = 1e-03$ and $n_iter = 100$ (left) with the ground truth point cloud (right) on the *bark* dataset with two original images presented (bottom).

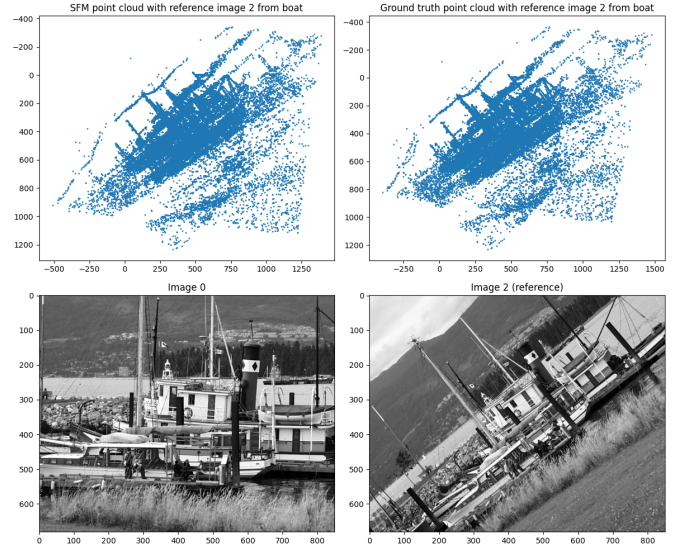


Fig. 2. Structure from motion estimated point cloud with parameters $\text{thresh} = 1e-05$ and $n_iter = 1000$ (left) with the ground truth point cloud (right) on the *boat* dataset with two original images presented (bottom).

have a very high inlier threshold when compared with the well-performing estimations. This suggests the poor homographies occur because large outliers are included within the inlier threshold when they are in fact outliers, therefore skewing the result to incorrect homographies.

B. Bundle Adjustment

In order to be as comprehensive as possible in the experimental evaluation of the implemented pipeline, a grid-search across a few of the hyper-parameters to the Gauss-Newton optimization was performed where the average residual between the ground truth points and the structure from motion

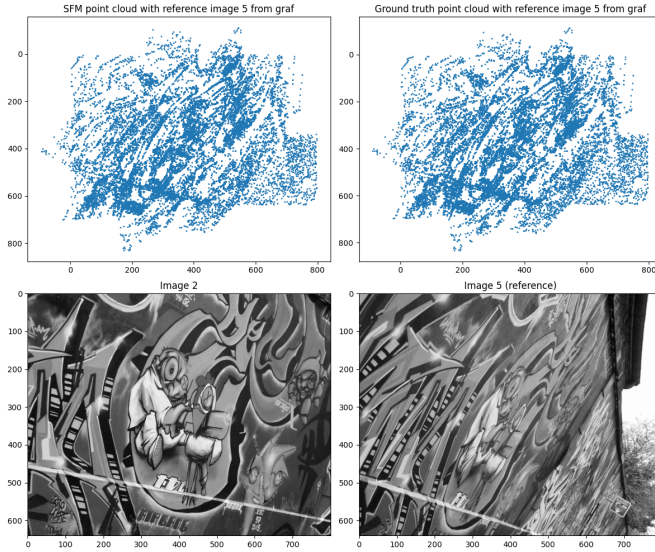


Fig. 3. Structure from motion estimated point cloud with parameters $\text{thresh} = 1e-05$ and $n_iter = 1000$ (left) with the ground truth point cloud (right) on the *graf* dataset with two original images presented (bottom).

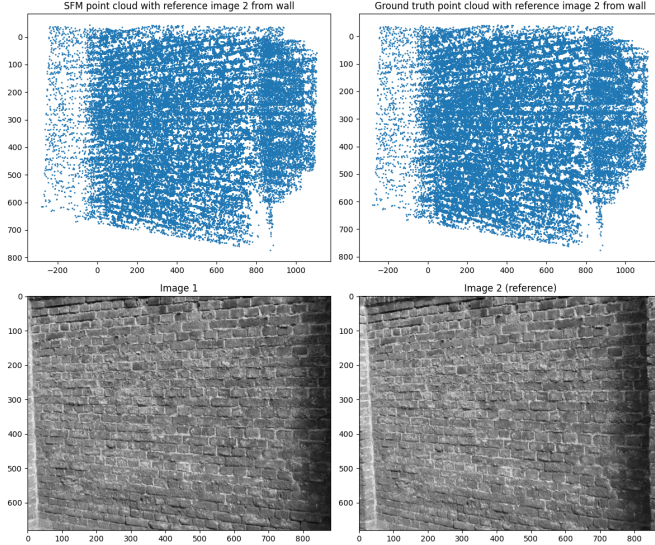


Fig. 4. Structure from motion estimated point cloud with parameters $\text{thresh} = 1e-05$ and $n_iter = 100$ (left) with the ground truth point cloud (right) on the *wall* dataset with two original images presented (bottom).

estimated points was used as a quality metric. The results of this hyperparameter search are shown in Appendix C while the best results are summarized in Table II.

Observe the improvements following bundle adjustment in Table III compared to Table I.

As shown in the above table, bundle adjustment produces only marginal improvements over the basic pipeline on the *graf* dataset, while *bark* and *boat* show a small regression in quality. The comparison for *bark* is complicated by a differing RANSAC threshold between the two pipelines. In order to make the grid-search for the bundle adjustment feasible (curse

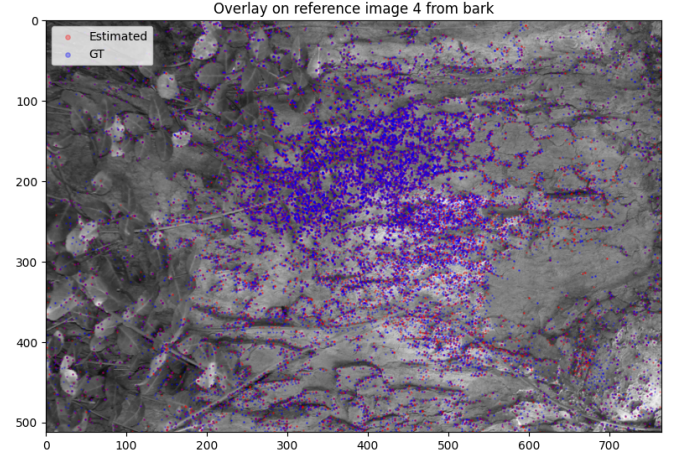


Fig. 5. Structure from motion estimated point cloud with parameters $\text{thresh} = 1e-04$ and $n_iter = 1000$ and ground truth point cloud overlaid on reference image of the *bark* dataset.

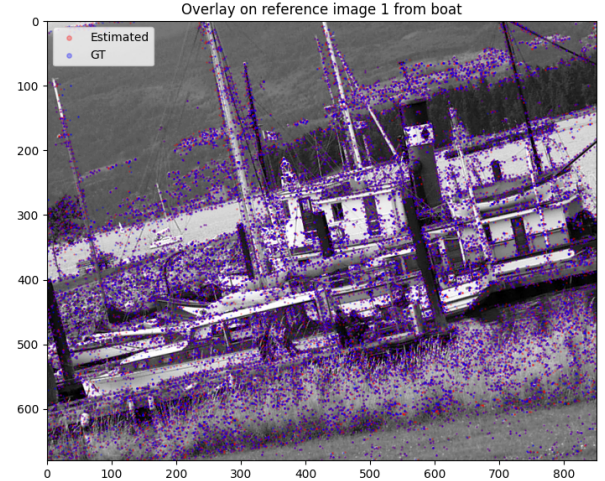


Fig. 6. Structure from motion estimated point cloud with parameters $\text{thresh} = 1e-04$ and $n_iter = 1000$ and ground truth point cloud overlaid on reference image of the *boat* dataset.

TABLE II
SUMMARY RESULTS OF THE BUNDLE ADJUSTMENT METHOD GRID SEARCH.

dataset	sample_ratio	gn_n_iter	avg_residual
bark	0.09	10.0	3.1512
boat	0.09	10.0	4.8097
graf	0.03	5.0	1.1676
wall	0.09	30.0	1.6331

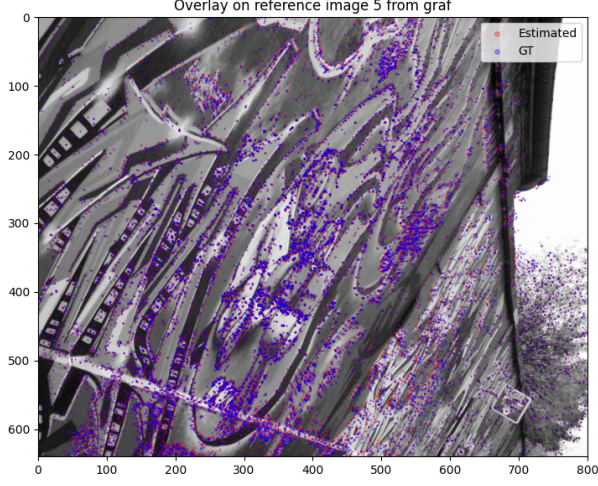


Fig. 7. Structure from motion estimated point cloud with parameters $\text{thresh} = 1e-04$ and $n_iter = 1000$ and ground truth point cloud overlayed on reference image of the *graf* dataset.

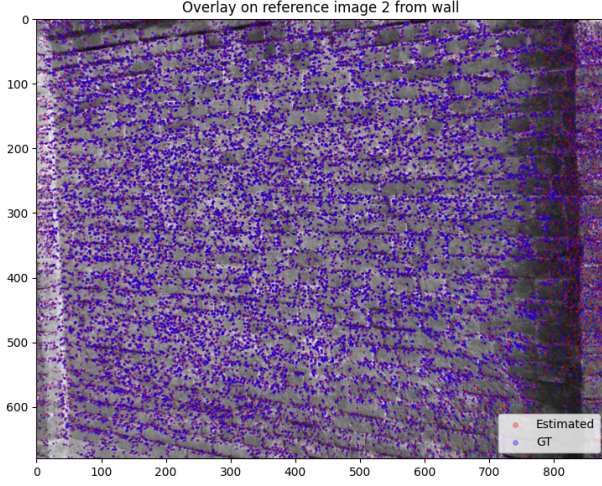


Fig. 8. Structure from motion estimated point cloud with parameters $\text{thresh} = 1e-04$ and $n_iter = 1000$ and ground truth point cloud overlayed on reference image of the *wall* dataset.

TABLE III
DIFFERENCE IN AVERAGE RESIDUAL BETWEEN ADVANCED AND BASIC METHODS PER DATASET (NEGATIVE = ADVANCED IS BETTER).

dataset	advanced	basic	difference
bark	3.1512	2.9697	0.1815
boat	4.8097	4.7630	0.0467
graf	1.1676	1.2388	-0.0712
wall	1.6331	1.5678	0.0653

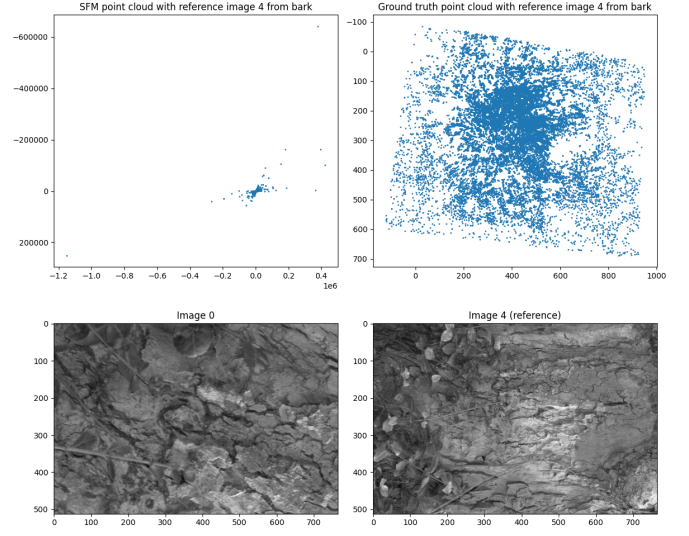


Fig. 9. Structure from motion estimated point cloud with parameters $\text{thresh} = 10$ and $n_iter = 10000$ (left) with the ground truth point cloud (right) on the *bark* dataset with two original images presented (bottom).

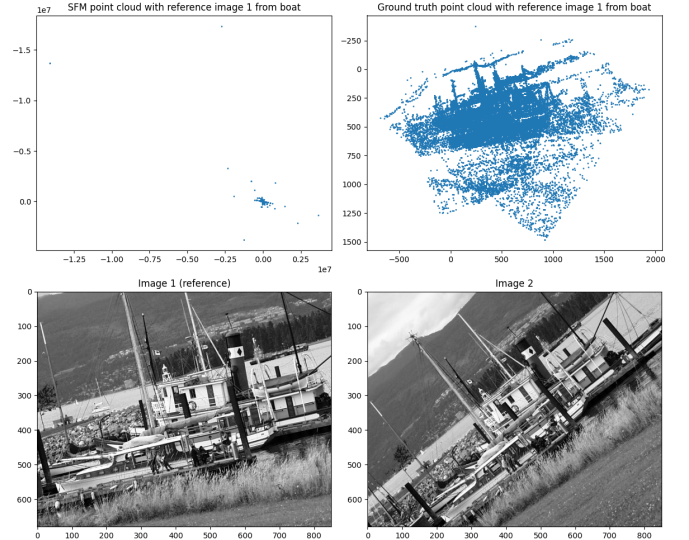


Fig. 10. Structure from motion estimated point cloud with parameters $\text{thresh} = 10$ and $n_iter = 100$ (left) with the ground truth point cloud (right) on the *boat* dataset with two original images presented (bottom).

of dimensionality applied to computation time) it was decided to set the RANSAC threshold and maximum number of iterations to $1e-05$ and 1000 respectively as this combination had the best results on the basic pipeline. Therefore, there is a large mismatch in the best parameters on the bark dataset as opposed to the other datasets which makes direct comparison incorrect. However, as shown in Appendix B, when using a threshold and number of iterations on the basic pipeline matching that for the grid-search on the bundle adjustment pipeline an average residual of ≈ 3.1973 is achieved which makes the true difference -0.0461 which is a slight improvement on par with

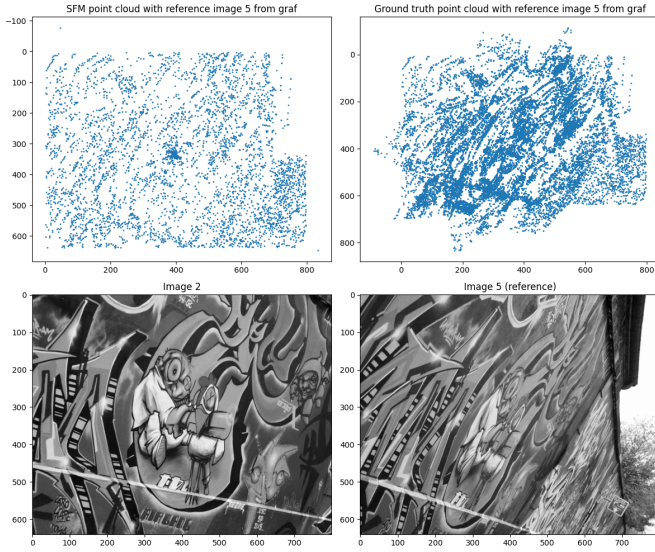


Fig. 11. Structure from motion estimated point cloud with parameters $\text{thresh} = 10$ and $n_{\text{iter}} = 10000$ (left) with the ground truth point cloud (right) on the *graf* dataset with two original images presented (bottom).

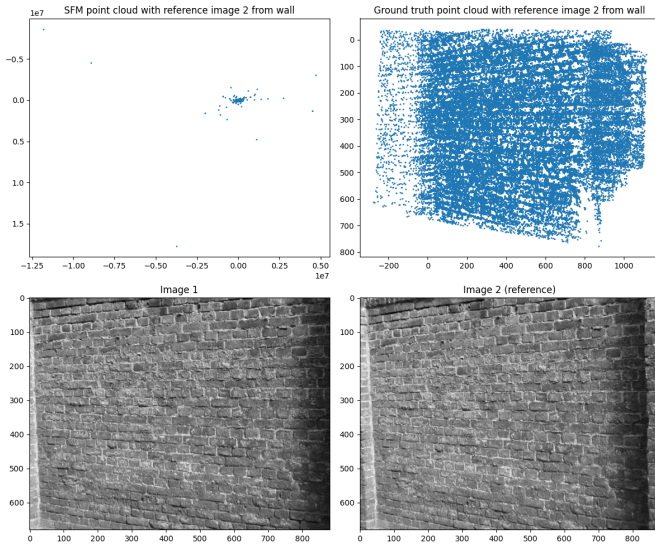


Fig. 12. Structure from motion estimated point cloud with parameters $\text{thresh} = 10$ and $n_{\text{iter}} = 10000$ (left) with the ground truth point cloud (right) on the *wall* dataset with two original images presented (bottom).

that for the *graf* dataset. Consequently, to accurately determine the improvement one must match the parameters exactly.

Nevertheless, even when all parameters line up the improvement is very small. These limited gains are attributed to two main factors. First, the grid search only tests up to 30 Gauss-Newton iterations, and many of the best configurations are found at this upper limit, suggesting the optimizer requires more iterations to make meaningful progress. Second, the stochastic mini-batch updates may introduce so much noise that the result is almost entirely dependent on getting good subsets during optimization. This is seen in the grid-search re-

sults where sometimes lower sample ratios outperform higher ones, suggesting that sample ratios between 0.03 and 0.12 are essentially equivalent in the result due to poor draws.

In future tests it would be desirable to test differing RANSAC thresholds and iterations alongside an expanded search over the Gauss-Newton hyperparameters. However, this would be incredibly costly and likely requires optimization of the implementation to increase performance. Another option is to use other statistical methods to find hyperparameters with a high likelihood of success.

In either case, it is clear from Table I that the range for improvement is already very low. For example, the datasets *graf* and *wall* have average residuals within 2 pixels which is near perfect before bundle adjustment. This also can affect the result since at this point step size on the Gauss-Newton updates comes into play. In order to optimize efficiently one must also take an appropriately sized step, especially when using a subset with mini-batches since it is much easier to overshoot the optimum. Perhaps a lower step size would have improved the results of the grid search, however with a lower step size more iterations are required to converge which would have increased the running time even more.

Overall, the results from the bundle adjustment extension to the basic pipeline are expected given the conditions for the problem. Given a small room for improvement, combined with an optimizer using only a small subset of the available data and a fixed step size, a large reduction in average residual was not expected. Instead, it is clear that the bundle adjustment algorithm is both functional and in some cases can lead to a small but measurable improvement in the result. Therefore, an improved implementation may be able to achieve greater results given more time.

The best result for each dataset is shown in Fig. 13, Fig. 14, Fig. 15, and Fig. 16 where it is clear in all cases that the estimated structure from motion point cloud matches the ground-truth point cloud within a very low error.

The worst result for each dataset is shown in Fig. 17, Fig. 18, Fig. 19, and Fig. 20.

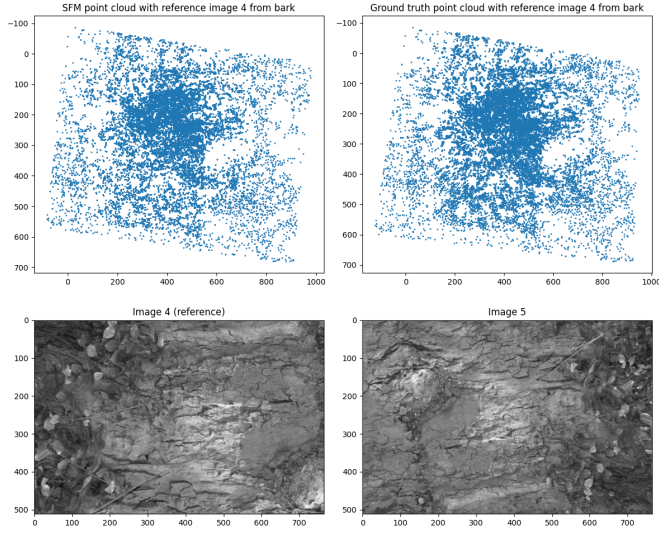


Fig. 13. Structure from motion estimated point cloud with parameters $\text{sample_ratio} = 0.09$ and $\text{gn_n_iter} = 10$ (left) with the ground truth point cloud (right) on the *bark* dataset with two original images presented (bottom).

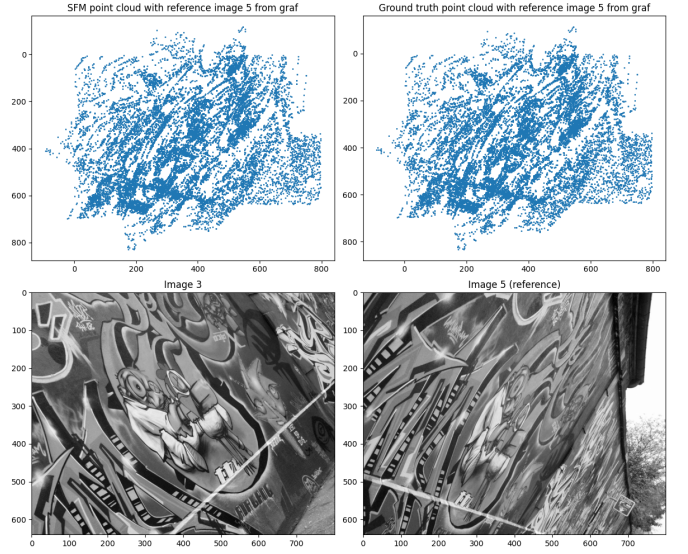


Fig. 15. Structure from motion estimated point cloud with parameters $\text{sample_ratio} = 0.03$ and $\text{gn_n_iter} = 5$ (left) with the ground truth point cloud (right) on the *graf* dataset with two original images presented (bottom).

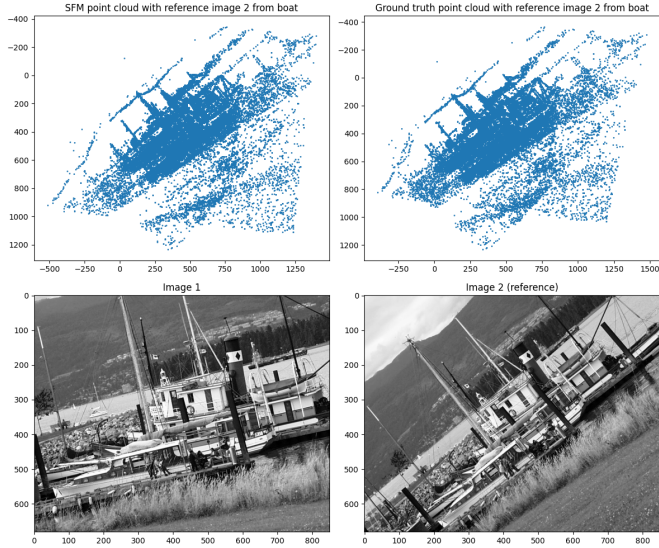


Fig. 14. Structure from motion estimated point cloud with parameters $\text{sample_ratio} = 0.09$ and $\text{gn_n_iter} = 10$ (left) with the ground truth point cloud (right) on the *boat* dataset with two original images presented (bottom).

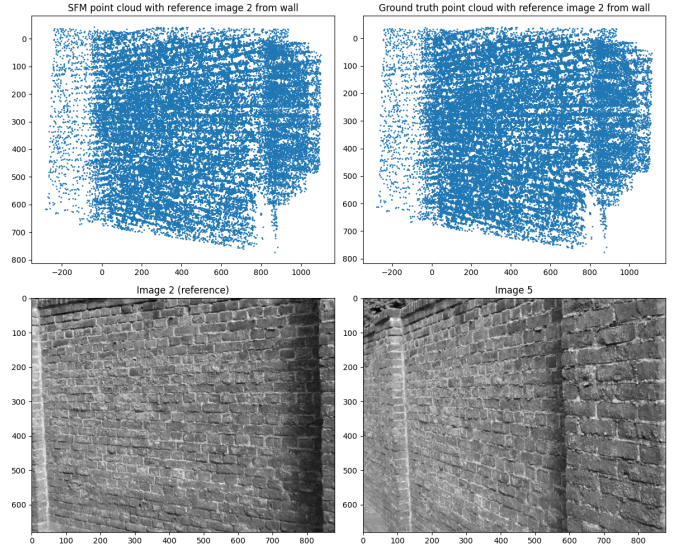


Fig. 16. Structure from motion estimated point cloud with parameters $\text{sample_ratio} = 0.09$ and $\text{gn_n_iter} = 30$ (left) with the ground truth point cloud (right) on the *wall* dataset with two original images presented (bottom).

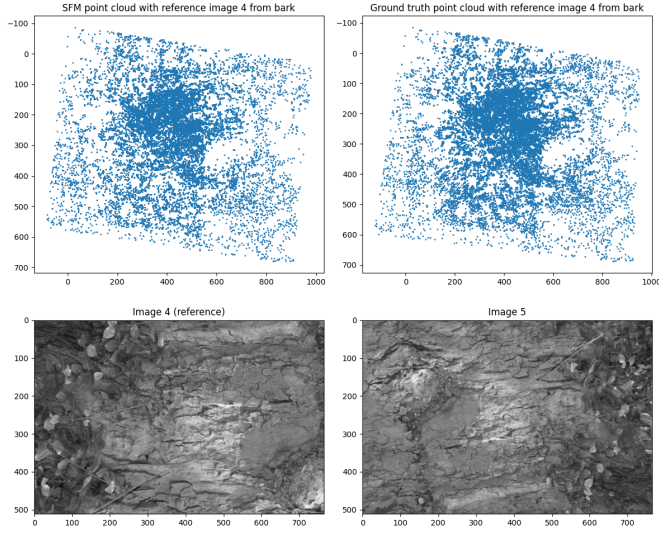


Fig. 17. Structure from motion estimated point cloud with parameters $\text{sample_ratio} = 0.12$ and $\text{gn_n_iter} = 5$ (left) with the ground truth point cloud (right) on the *bark* dataset with two original images presented (bottom).

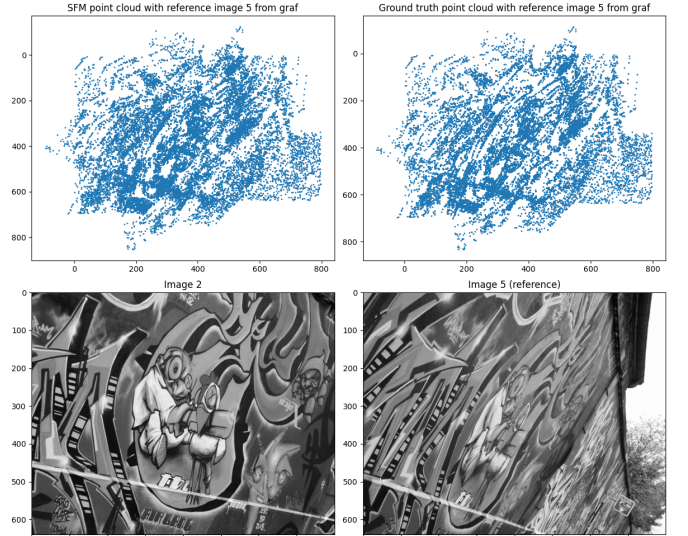


Fig. 19. Structure from motion estimated point cloud with parameters $\text{sample_ratio} = 0.03$ and $\text{gn_n_iter} = 30$ (left) with the ground truth point cloud (right) on the *graf* dataset with two original images presented (bottom).

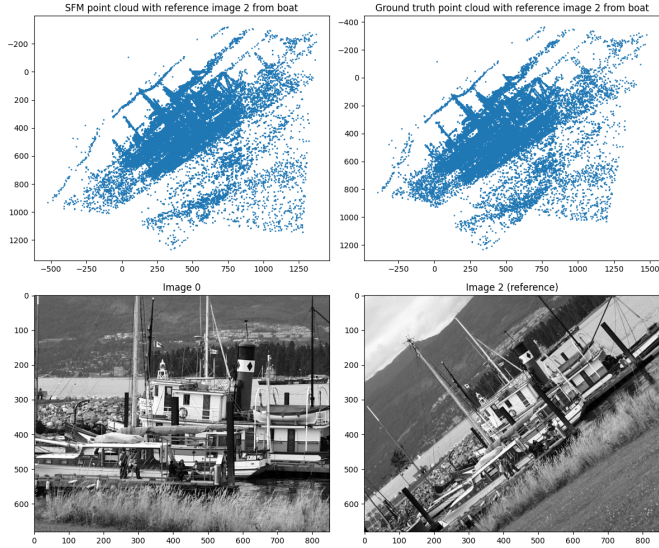


Fig. 18. Structure from motion estimated point cloud with parameters $\text{sample_ratio} = 0.12$ and $\text{gn_n_iter} = 10$ (left) with the ground truth point cloud (right) on the *boat* dataset with two original images presented (bottom).

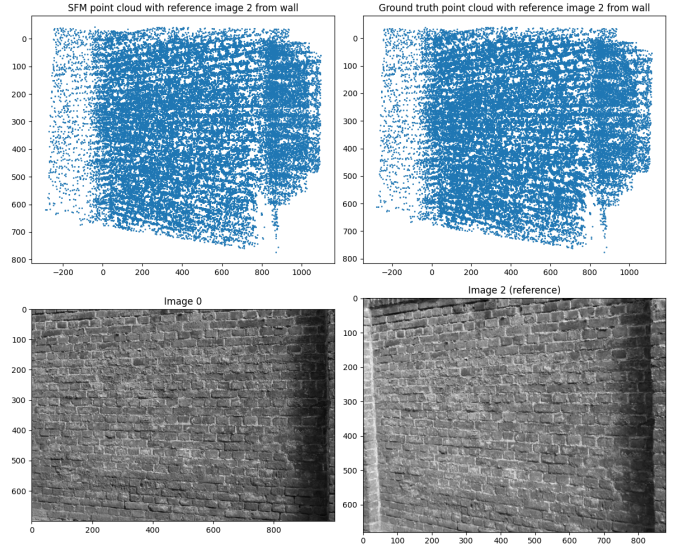


Fig. 20. Structure from motion estimated point cloud with parameters $\text{sample_ratio} = 0.12$ and $\text{gn_n_iter} = 10$ (left) with the ground truth point cloud (right) on the *wall* dataset with two original images presented (bottom).

APPENDIX A STATEMENT ON THE USAGE OF AI

Claude was used as a reference and debugging tool throughout this project to clarify mathematical concepts, identify implementation bugs, and assist with report writing. All implementations were written independently, with the AI serving as an explanatory resource and reviewer. All reasoning, analysis, and conclusions in this report are my own.

Specific uses include:

- Clarifying the DLT algorithm for homography estimation and the role of similarity transforms for numerical conditioning.
- How to select an appropriate RANSAC inlier threshold relative to normalized coordinate space.
- Understanding the union-find data structure for building feature tracks from pairwise inlier correspondences.
- Clarifying the Gauss-Newton bundle adjustment formulation including parameter vector layout, residual computation, finite difference Jacobian approximation, and subset sampling strategy.
- Debugging numpy issues including incorrect use of `np.flatten`, `hstack` as method vs function, and singular matrix errors in `np.linalg.solve`.
- Clarifying the reprojection residual direction. Applying `inv(H_I)` to map global points back into image space for comparison with observed keypoints.
- Looking up Python and numpy syntax including set operations, `itertools.combinations/product`, tuple dict keys, and array slicing.
- Explaining the Gauss-Newton method and how solving the overdetermined linear system $J\delta = r$ via least-squares implicitly computes the update $(J^T J)^{-1} J^T r$ without explicitly forming the normal equations.
- Identifying a bug in the similarity transform where the average norm was computed from raw (uncentered) point coordinates instead of centered coordinates, causing incorrect DLT normalization scaling.
- Identifying that the fixed evaluation subset for early stopping must be sampled once before the optimization loop rather than per iteration, as per-iteration resampling makes residuals across iterations incomparable.
- Discussing why bundle adjustment shows limited improvement over the basic pipeline, specifically the role of insufficient iterations, stochastic noise in mini-batch updates, and the mismatch between the optimization objective and the evaluation metric.
- Reviewing the report method section for accuracy against the implementation and identifying discrepancies.
- Explaining LaTeX pseudocode syntax using the algorithmic and algorithm packages, including the input/output declarations and set-builder notation used in the algorithm blocks.
- Generating small test cases for some methods in order to verify correctness of implementation (Feature tracks, RANSAC estimate, DLT, etc.).

APPENDIX B BASIC METHOD GRID SEARCH

Grid search on RANSAC hyperparameters evaluated by average residuals on ground truth homographies. Per column, the best (lowest) value is highlighted green and the worst (highest) red for *avg_residual*.

TABLE IV
GRID SEARCH RESULTS, DATASET = BARK, N_ITER = 100.

thresh	n_iter	avg_residual
10.0	100	21.7550
1.0	100	15.2757
0.1	100	3.8865
0.01	100	4.2638
0.001	100	2.9697
0.0001	100	3.1573
1e-05	100	3.1937

TABLE V
GRID SEARCH RESULTS, DATASET = BARK, N_ITER = 1000.

thresh	n_iter	avg_residual
10.0	1000	44.5494
1.0	1000	11.4601
0.1	1000	3.9690
0.01	1000	3.6792
0.001	1000	3.2531
0.0001	1000	3.1726
1e-05	1000	3.1973

TABLE VI
GRID SEARCH RESULTS, DATASET = BARK, N_ITER = 10000.

thresh	n_iter	avg_residual
10.0	10000	686.1299
1.0	10000	25.5223
0.1	10000	4.5816
0.01	10000	3.3126
0.001	10000	3.3604
0.0001	10000	3.1527
1e-05	10000	3.1866

TABLE VII
GRID SEARCH RESULTS, DATASET = BOAT, N_ITER = 100.

thresh	n_iter	avg_residual
10.0	100	4609.9400
1.0	100	38.6722
0.1	100	20.7595
0.01	100	9.3246
0.001	100	6.4974
0.0001	100	6.4364
1e-05	100	5.4394

TABLE VIII
GRID SEARCH RESULTS, DATASET = BOAT, N_ITER = 1000.

thresh	n_iter	avg_residual
10.0	1000	456.8946
1.0	1000	46.0945
0.1	1000	12.2141
0.01	1000	7.0105
0.001	1000	6.4918
0.0001	1000	6.6302
1e-05	1000	4.7630

TABLE IX
GRID SEARCH RESULTS, DATASET = BOAT, N_ITER = 10000.

thresh	n_iter	avg_residual
10.0	10000	296.6585
1.0	10000	90.4336
0.1	10000	38.5844
0.01	10000	9.2585
0.001	10000	6.9618
0.0001	10000	5.8808
1e-05	10000	5.0702

TABLE X
GRID SEARCH RESULTS, DATASET = GRAF, N_ITER = 100.

thresh	n_iter	avg_residual
10.0	100	128.9233
1.0	100	40.6648
0.1	100	11.3335
0.01	100	4.2105
0.001	100	1.5512
0.0001	100	1.9587
1e-05	100	1.5649

TABLE XI
GRID SEARCH RESULTS, DATASET = GRAF, N_ITER = 1000.

thresh	n_iter	avg_residual
10.0	1000	165.5615
1.0	1000	44.9959
0.1	1000	12.6143
0.01	1000	2.4562
0.001	1000	1.6697
0.0001	1000	1.4823
1e-05	1000	1.2388

TABLE XII
GRID SEARCH RESULTS, DATASET = GRAF, N_ITER = 10000.

thresh	n_iter	avg_residual
10.0	10000	172.3440
1.0	10000	64.6876
0.1	10000	10.2845
0.01	10000	4.5699
0.001	10000	1.8063
0.0001	10000	1.6003
1e-05	10000	1.2548

TABLE XIII
GRID SEARCH RESULTS, DATASET = WALL, N_ITER = 100.

thresh	n_iter	avg_residual
10.0	100	127.5900
1.0	100	26.4458
0.1	100	9.5649
0.01	100	5.2535
0.001	100	2.6534
0.0001	100	1.6098
1e-05	100	1.5678

TABLE XIV
GRID SEARCH RESULTS, DATASET = WALL, N_ITER = 1000.

thresh	n_iter	avg_residual
10.0	1000	2236.8699
1.0	1000	70.6886
0.1	1000	9.8424
0.01	1000	3.6111
0.001	1000	3.6332
0.0001	1000	1.7079
1e-05	1000	1.7265

TABLE XV
GRID SEARCH RESULTS, DATASET = WALL, N_ITER = 10000.

thresh	n_iter	avg_residual
10.0	10000	3128.5872
1.0	10000	90.0053
0.1	10000	14.6782
0.01	10000	4.4612
0.001	10000	3.6633
0.0001	10000	1.8429
1e-05	10000	1.7033

APPENDIX C

ADVANCED METHOD GRID SEARCH

Grid search on *sample_ratio* and Gauss-Newton max iterations evaluated by average residuals on ground truth homographies. Per column, the best (lowest) value is highlighted green and the worst (highest) red for *avg_residual*.

TABLE XVI

GRID SEARCH RESULTS, DATASET = BARK, GN_N_ITER = 5.0.

sample_ratio	gn_n_iter	avg_residual
0.03	5.0	3.1744
0.06	5.0	3.1699
0.09	5.0	3.1823
0.12	5.0	3.1992

TABLE XVII

GRID SEARCH RESULTS, DATASET = BARK, GN_N_ITER = 10.0.

sample_ratio	gn_n_iter	avg_residual
0.03	10.0	3.1954
0.06	10.0	3.1734
0.09	10.0	3.1512
0.12	10.0	3.1837

TABLE XVIII

GRID SEARCH RESULTS, DATASET = BARK, GN_N_ITER = 30.0.

sample_ratio	gn_n_iter	avg_residual
0.03	30.0	3.1866
0.06	30.0	3.1775
0.09	30.0	3.1745
0.12	30.0	3.1794

TABLE XIX

GRID SEARCH RESULTS, DATASET = BOAT, GN_N_ITER = 5.0.

sample_ratio	gn_n_iter	avg_residual
0.03	5.0	5.1864
0.06	5.0	5.2186
0.09	5.0	5.2552
0.12	5.0	5.5056

REFERENCES

- [1] R. Hartley and A. Zisserman, *Multiple View Geometry in Computer Vision*, 2nd ed. Cambridge: Cambridge University Press, 2004.
- [2] "Disjoint-set data structure," Wikipedia, The Free Encyclopedia. [Online]. Available: https://en.wikipedia.org/wiki/Disjoint-set_data_structure. [Accessed: May 30, 2026].

TABLE XX

GRID SEARCH RESULTS, DATASET = BOAT, GN_N_ITER = 10.0.

sample_ratio	gn_n_iter	avg_residual
0.03	10.0	5.2475
0.06	10.0	4.9707
0.09	10.0	4.8097
0.12	10.0	6.2909

TABLE XXI

GRID SEARCH RESULTS, DATASET = BOAT, GN_N_ITER = 30.0.

sample_ratio	gn_n_iter	avg_residual
0.03	30.0	4.8721
0.06	30.0	4.9889
0.09	30.0	4.9211
0.12	30.0	4.8968

TABLE XXII

GRID SEARCH RESULTS, DATASET = GRAF, GN_N_ITER = 5.0.

sample_ratio	gn_n_iter	avg_residual
0.03	5.0	1.1676
0.06	5.0	1.3681
0.09	5.0	1.5766
0.12	5.0	1.6125

TABLE XXIII

GRID SEARCH RESULTS, DATASET = GRAF, GN_N_ITER = 10.0.

sample_ratio	gn_n_iter	avg_residual
0.03	10.0	4.5722
0.06	10.0	3.3014
0.09	10.0	1.8167
0.12	10.0	2.7365

TABLE XXIV

GRID SEARCH RESULTS, DATASET = GRAF, GN_N_ITER = 30.0.

sample_ratio	gn_n_iter	avg_residual
0.03	30.0	5.6456
0.06	30.0	2.1145
0.09	30.0	1.8082
0.12	30.0	1.9121

TABLE XXV

GRID SEARCH RESULTS, DATASET = WALL, GN_N_ITER = 5.0.

sample_ratio	gn_n_iter	avg_residual
0.03	5.0	1.6933
0.06	5.0	1.6951
0.09	5.0	1.6512
0.12	5.0	1.7404

TABLE XXVI
GRID SEARCH RESULTS, DATASET = WALL, GN_N_ITER = 10.0.

sample_ratio	gn_n_iter	avg_residual
0.03	10.0	1.6962
0.06	10.0	1.6571
0.09	10.0	1.7236
0.12	10.0	1.7467

TABLE XXVII
GRID SEARCH RESULTS, DATASET = WALL, GN_N_ITER = 30.0.

sample_ratio	gn_n_iter	avg_residual
0.03	30.0	1.6558
0.06	30.0	1.7061
0.09	30.0	1.6331
0.12	30.0	1.6657